

Lab #7: Games with Greenfoot

You can work individually or with a partner and make one submission for both of you.

If you work with a partner, make sure your schedules are compatible, so that you can genuinely work together. Don't split up the work; do it together at the same time.

Install Greenfoot

<https://www.greenfoot.org/download>

It runs on both Windows, Mac, and Linux.

Check out the example games

4 examples are attached to this lab.

- Hedgehog
- Shooting Game (where Nick P shoots butterflies with his face)
- Cards
- Zelda3 (move between screens, basic animation, simple enemies, simple dialog window)

There are also videos about each project! Check Classroom for more details (it's posted there)

Build a game! Here are some ideas

You cannot make a version of the following games

- Flappy Bird
- Snake game

This is because there are dozens of different versions of these available online. Pick something different and distinct.

Pong



Two paddles that you can move with different keys on the keyboard. The ball bounces around when it hits the edge. If the ball gets past a paddle, that scores a point

RPG

Travel around between screens gathering up coins or resources or items, interacting with other actors, and solving simple puzzles. The puzzles can be things like "get the key from three screens over and use it to open the treasure chest". Having the key could just be that your keys instance variable is greater than 0, the treasure chest could be an actor, and when you open it, your treasure instance variable could be incremented by 1.

Space Invaders



Move your ship along the bottom of the screen, and fire missiles at enemies. The enemies slowly move down towards the player's ship.

Your game should be different than the examples

Don't make minor tweaks to the example games. That's boring! Be interesting.

How to use Custom Images

Check out Zelda3! There are custom images in the images folder.

If you put images into the images folder inside your Greenfoot project (use Finder on Mac or Explorer/Navigator/whatever it's called on Windows) to do this.

To use your custom image in `YourActor`, you would put something like this in the constructor:

```
public YourActor() {  
    GreenfootImage image = new GreenfootImage("images/yourfile.png");  
    setImage(image);  
}
```

Where to get new images

Check out itch.io! This is where indie game devs put sample games, and also sample art assets. Note that "assets" is the general term that the game industry uses for artwork, such as buttons, icons, sprites (characters), and so on.

Some of the assets on itch have a cost, but many have a "suggested" cost and allow you to download and use them for free.

However, if you download and use something, you must credit the author! Please keep track of what art assets you download (just copy the link from itch).

It's classy to credit creators!

What to submit

- Zip your project folder and upload to classroom.
Please comment who your partner is and which code submission we should grade.

- Include a ***video*** or ***Google doc*** writeup describing 3 interesting or challenging features you implemented. This can be short. **Each person should upload their own video.**
- Fill out the lab report.